

Smart Dock Assistant

A hands-free helper that tells you, through your headphones, which Citi Bike dock near your destination actually has space, so you never have to pull out your phone while riding.

The problem

Riding Citi Bikes to class, we kept arriving to find the nearby docks already full. Then the scramble started: stop, unlock the phone, open the Citi Bike app, hunt for another station. That's annoying and unsafe on a bike.

We wanted something that checks live dock availability for you, remembers the docks you normally use, and reroutes you before you get there, entirely by voice.

Three ways to use it

AUTOMATIC

Arrival alert

no input needed

Get within ~500 m of school, work or home and your phone speaks up on its own: is your usual dock open, and if not, where to go instead.

VOICE

Dock Finder

"Hey Siri, run Dock Finder"

Siri asks what you need. Ask about any station, any place, or "a dock near me". Your GPS location goes along with every question.

TYPED

Chat

n8n chat window

The original version: a text chat that gives fuller answers with Google Maps cycling directions. Handy for planning and testing.

What it actually says

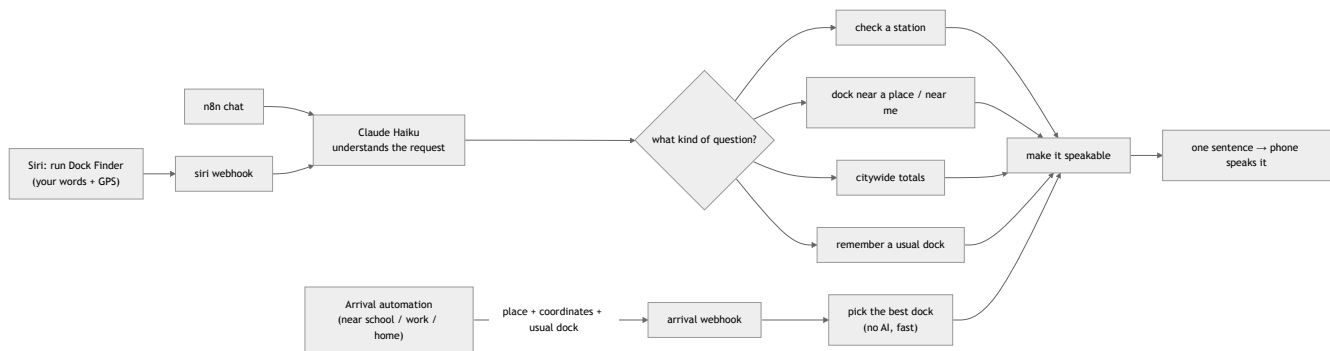
Answers are short on purpose, one sentence you can catch at a red light. These are real responses from live Citi Bike data:

USUAL DOCK OK	“your usual dock has 25 docks open.”
USUAL DOCK FULL	“your usual dock is full. go to Washington and Greene, 8 docks, 100 meters from school.”
NEAR ME	“West 15th and 6th has 53 docks open, right there.”
STATION CHECK	“Mercer and Bleecker has 29 docks open.”
NEAR A PLACE	“East 17th and Broadway has 47 docks open, 150 meters from union square.”
WHOLE SYSTEM	“35,393 bikes and 30,882 open docks citywide. 243 stations are full.”

Station names are turned into how New Yorkers say them (“W 15 St & 6 Ave” becomes “West 15th and 6th”), so Siri doesn't read “St” as “Saint”.

How it works

The iPhone does the listening and the talking; an n8n workflow in the cloud does the thinking. Every request ends in a single sentence that the phone reads aloud with its built-in voice.



- **Live data:** Citi Bike's public real-time feed (GBFS), which needs no API key. A station counts as available only if it's accepting returns and has at least 2 open docks, within 1.2 km.
- **Place lookup:** OpenStreetMap turns “NYU Stern” or “Union Square” into coordinates.
- **Understanding requests:** a small, fast AI model (Claude Haiku 4.5) classifies what you said into one of six request types and pulls out station names and places. It never writes the answer itself; plain code does, so the numbers are always straight from the feed.
- **Memory:** the Siri conversation remembers aliases like “my school dock”. Arrival alerts carry their usual dock inside the automation.

Decisions we made along the way

Trigger near saved places, not the Google Maps destination

Google Maps doesn't let other apps see your navigation destination or live position. Since we mostly commute to the same few places, iPhone “arrive” automations for school, work and home give us a fully automatic trigger without building an app.

Siri + Shortcuts instead of Telegram

Our first voice version used a Telegram bot, but sending a voice note means unlocking the phone and holding a button, and replies don't auto-play. That breaks the no-touch rule. Siri is truly hands-free, and the phone handles speech-to-text and text-to-speech for free.

Short, spoken-first answers

The chat keeps detailed answers with direction links, but anything spoken is trimmed to one sentence: where to go, how many docks, how far.

AI only where it helps

Arrival alerts have fixed inputs, so they skip the AI entirely and come back in 2–3 seconds. Only free-form Siri questions go through the model (3–5 seconds).

Running costs

PIECE	COST
Citi Bike live data, OpenStreetMap lookups, iPhone speech	Free
AI model per Siri question	≈ \$0.002 (arrival alerts use none)
n8n Cloud hosting	Our current plan; could move to a free Cloudflare Worker

Limits and what's next

- Arrival alerts fire however you arrive, including on foot or by subway. iOS can't tell you're on a bike, so a commute time window is the only filter.
- Distances are straight-line, not riding distance.
- Siri conversation memory resets if the n8n server restarts.
- **Next:** move the backend off n8n to a free serverless function that caches the station list for faster answers; keep each rider's places on the server so automations need just one action; and eventually a native app that sets up the arrival alert for *any* destination you share from Google Maps.

Set it up on your phone

Grab the setup handoff file from the group chat (`citibike-dock-assistant-iphone-setup-HANDOFF.md`). Attach it to ChatGPT or Claude and say “help me set this up on my iPhone”. It walks you through the arrival automations and the Dock Finder shortcut step by step, including every mistake we hit along the way. You'll need iOS 17 or newer and about 15 minutes.

Built with n8n, the iPhone Shortcuts app, Citi Bike's public GBFS feed, OpenStreetMap, and Claude Haiku 4.5. Not affiliated with Citi Bike or Lyft.